

Department of Mathematics and Applied Mathematics
3DM: Discrete Mathematics

Test 1

Noon 12 May – Noon 13 May 2020

50 marks = 100%



- This question paper consists of **6 pages** (including this one).
- You must upload your solution to the Assignments Tool in the 3DM Vula no later than noon Wednesday 13 May.
- **To make things simpler for the markers, please upload your solution as a PDF if possible.**
- Please put your name and student number on your solution.
- You don't have to do the questions in the same order as they are below. But please number them clearly.
- In Question 1, you must just give the answer (i.e., no proofs/working necessary). In Question 2, you must show all working/proofs (no working/proof = no marks, even if your answer is correct).
- You may type your solution (I strongly suggest if you do that you use $\text{\LaTeX}$) or (if you have a tablet) write on this question paper and upload the PDF.
- Or you may write your solution out and upload photographs/scans of the pages. If you do this, please make sure each page is clearly legible. **If the photograph is not clear or if we can't read your handwriting, we will mark your solution incorrect.**
- The mathematics in this test should only take 1 – 2 hours to finish. The rest of the time between noon Tuesday and noon Wednesday is to accommodate difficulties connecting etc. You should therefore finish the test as soon as possible and upload your solution as soon as you are finished. **Don't wait till the last minute to upload your solution!**

OPEN BOOK TEST RULES

This is an Open Book test. That means:

- You may refer to the course notes and your homework answers.
- You may not use the internet to get answers or help. I had to think about this one for a while. Not everyone in the class can easily access the internet. Allowing students to cruise around the internet looking for things to help them with a test they're busy writing risks creating a very uneven playing field.
- You may not communicate with anyone about the questions (that includes other students in the course, people on the internet, your grandmother the maths professor, your friend who did the course 3 years ago, etc).

Please read the Faculty of Science Student Honour Code, below.

FACULTY OF SCIENCE STUDENT HONOUR CODE

The University of Cape Town's official statement of general rules and policies, including a statement of values and expected student and staff conduct, can be found [here](#). All staff and students in the Faculty of Science are required to familiarise themselves with UCT's rules and policies. The Faculty of Science takes UCT's statement of values seriously. Among other things, every member of the Faculty of Science — student and staff — is expected to strive for personal and academic integrity; to be respectful and tolerant towards others; to honour the rights, personal choices and property of others; to take responsibility for their behaviour individually and in groups; and generally to act as a responsible citizen in our academic community and beyond. As stated in the above mentioned document: "Truth, fairness, consistency and integrity in both academic and other work, and in all personal and institutional relationships" are key elements of UCT's academic values. We are currently in a global crisis, and recognise that distance learning will be a challenge for many students. Nonetheless, we must strive to uphold these values. We ask students to commit to an Honour Code — i.e. doing what is morally right — when completing tasks remotely. In particular, all assignments and tests must be a student's own work, with no input from others (unless otherwise specified), and no work can be plagiarised. We also expect students to report to the Faculty if they believe that other UCT students have breached this code. The Faculty of Science supports our students in this difficult time, and requests their full commitment to the integrity of the academic project. Together we will persevere, and succeed.

Question 1. [32 points] In this question (Question 1), just give the answer. Do not give proofs or working.

a. Give an example of each of the following (if such a graph exists) or write Does Not Exist (if it doesn't).

(i) A graph of order 4 that is not an induced subgraph of K_4 . (3)

P_4 is one such graph.

(ii) A graph of order 7 in which every induced subgraph is complete. (3)

K_7 is the only such graph.

(iii) A graph of size 8 in which no two vertices have the same degree. (3)

Does Not Exist (see HW Question 1.5).

(iv) A graph of order 12 that has six even vertices and six odd vertices. (3)

One such graph is $6K_1 \cup 3K_2$.

(v) A digraph of order 10 in which every vertex has indegree 2 and outdegree 3. (3)

Does Not Exist (see HW Question 1.13).

(vi) A graph with degree sequence 5, 5, 5, 5, 3, 3, 3, 3, 1. (3)

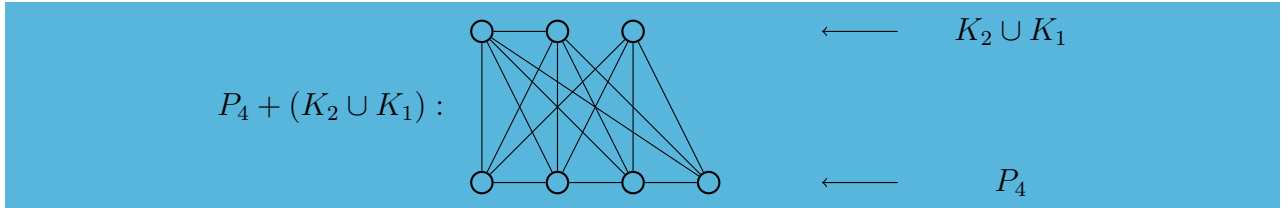
Does Not Exist (by Corollary 2).

b. Draw the following graphs.

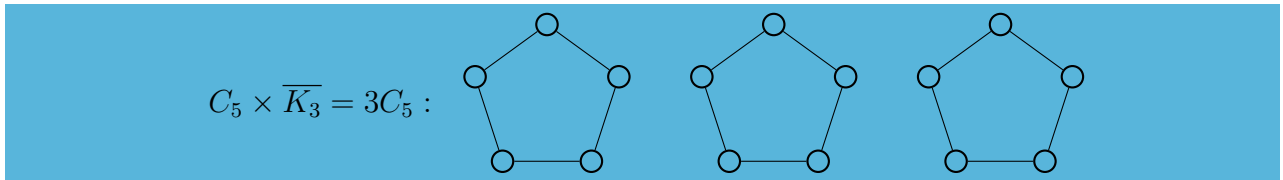
(i) $\overline{C_4}$. (2)



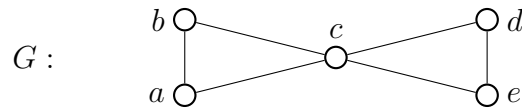
(ii) $P_4 + (K_2 \cup K_1)$. (2)



(iii) $C_5 \times \overline{K_3}$. (2)



c. Determine all automorphisms of the graph



Write each automorphism in cycle notation. (8)

There are 8 automorphisms:

- (1) $(a)(b)(c)(d)(e)$ (the identity)
- (2) $(ab)(c)(d)(e)$
- (3) $(ab)(c)(de)$
- (4) $(a)(b)(c)(de)$
- (5) $(ad)(be)(c)$
- (6) $(ae)(bd)(c)$
- (7) $(adbe)(c)$
- (8) $(aebd)(c)$

Question 2. [20 points] In this question (Question 2), you must show all proofs/working. No proofs/working = no marks.

- a. Prove or disprove: If G and H are graphs, then $\overline{G + H} = \overline{G} + \overline{H}$. (5)

This is false. To see this, consider the following counterexample: Let $G = H = K_1$. Then

$$\overline{G + H} = \overline{K_1 + K_1} = \overline{K_2}$$

while

$$\overline{G} + \overline{H} = \overline{K_1} + \overline{K_1} = K_1 + K_1 = K_2$$

Since $K_2 \neq \overline{K_2}$, this disproves the statement.

- b. Prove or disprove: If G and H are graphs, then

$$m(G \times H) = m(G)n(H) + n(G)m(H).$$

(5)

This is true.

Proof. From the notes, we have $E(G \times H) = \{(g_1, h_1)(g_2, h_2) : \text{(i) } g_1g_2 \in E(G) \text{ and } h_1 = h_2, \text{ or, (ii) } g_1 = g_2 \text{ and } h_1h_2 \in E(H)\}$. Let's define

$$\begin{aligned} E_1 &= \{(g_1, h_1)(g_2, h_2) : g_1g_2 \in E(G) \text{ and } h_1 = h_2\} \\ &= \{(g_1, v)(g_2, v) : g_1g_2 \in E(G) \text{ and } v \in V(H)\} \end{aligned}$$

and

$$\begin{aligned} E_2 &= \{(g_1, h_1)(g_2, h_2) : g_1 = g_2 \text{ and } h_1h_2 \in E(H)\} \\ &= \{(v, h_1)(v, h_2) : v \in V(G) \text{ and } h_1h_2 \in E(H)\}. \end{aligned}$$

Then $E(G \times H) = E_1 \cup E_2$; since $E_1 \cap E_2 = \emptyset$, we get

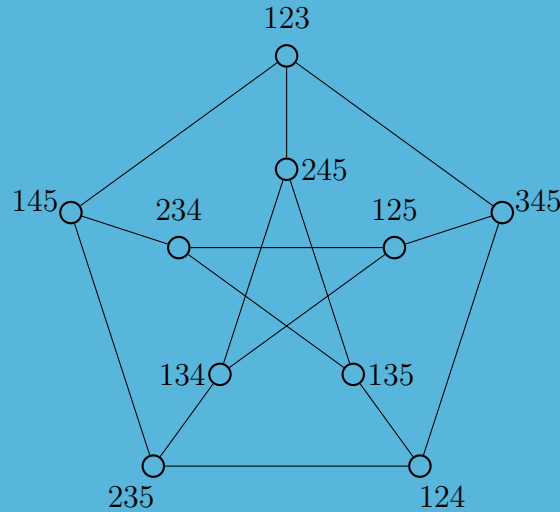
$$m(G \times H) = |E_1| + |E_2|.$$

For every vertex $v \in V(H)$, there are $m(G)$ edges of the form $(g_1, v)(g_2, v)$ in $G \times H$, so $|E_1| = m(G)n(H)$. Similarly, $|E_2| = n(G)m(H)$. The result follows. $\square$

- c. For an integer $t \geq 3$, we define $G(t)$ as the graph whose vertices are the 3-subsets of the set $S = \{1, 2, \dots, t\}$, and where two vertices u and v are adjacent if $|u \cap v| = 1$.

- (i) Draw the graph $G(5)$ (no working/proof needed). (3)

$G(5)$ is the Petersen Graph. For convenience, in the graph below the set $\{a, b, c\}$ is denoted abc .



- (ii) Determine the order of $G(t)$. (2)

The order of $G(t)$ is the number of 3-subsets of $S = \{1, 2, \dots, t\}$, which is by definition $\binom{t}{3}$.

- (iii) Determine the size of $G(t)$. (5)

Let $v = \{v_1, v_2, v_3\}$ be a vertex of $G(t)$. Each vertex adjacent to v contains one of the three elements v_1, v_2, v_3 and two other elements chosen from the $t - 3$ elements of $S - v$. It follows that

$$\deg v = 3 \binom{t-3}{2}.$$

Using the fact that $G(t)$ has order $\binom{t}{3}$, from Theorem 1 in the notes we have

$$m = \frac{1}{2} \sum_{v \in V(G(t))} \deg v = \frac{3}{2} \binom{t}{3} \binom{t-3}{2}$$

N.B: Whenever you work something out, you should do a ‘sanity check’ to make sure it makes sense. In this case, we can compare the number of vertices and edges in $G(5)$, which we worked out in (i), to the number of vertices predicted by our formulae in (ii) and (iii). If you do this, you’ll see that you get the same numbers.